Module 3 — DADA2 in Galaxy to phyloseq

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Module 3 — DADA2 in Galaxy to phyloseq

Goal: Process amplicon sequencing reads in Galaxy using DADA2, examine key outputs, then prepare the files needed for analysis in **phyloseq**.
Submission: Fill in answers, upload screenshots where prompted, then click **Prepare submission preview** and **Download as PDF (Print)**.

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0) Your dataset

0A. Sample set / project name:

WCSP microbiome

0B. Sequencing type:

16S V4 paired-end

0C. Metadata grouping variable you plan to analyze in phyloseq:

SampleSource

DADA2 workflow order

Use the Galaxy tools in this order:

```
plotQualityProfile → filterAndTrim → learnErrors → dada → mergePairs →
makeSequenceTable → removeBimeraDenovo → assignTaxonomy → sequence
counts
```

Reminder: The point of this workflow is to turn messy raw reads into a clean ASV table with taxonomy that can be analyzed ecologically in **phyloseq**.

Part 1 — Quality profiles

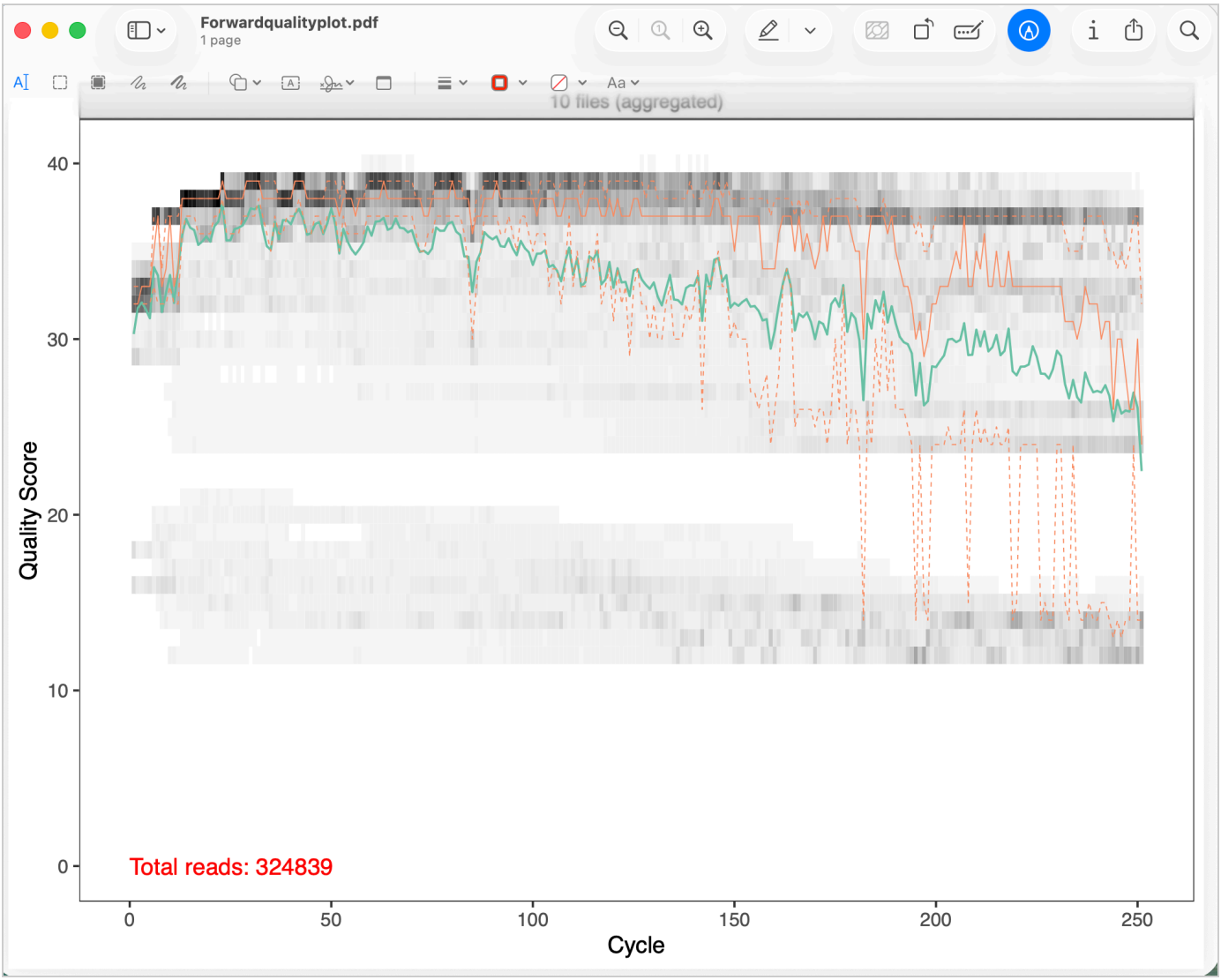
1A) Forward read quality profile

What is the purpose of this step?

The purpose is to analyze the length of the quality of the sequence, specifically allowing us to see how where the quality of the sequence drops and visualize where to trim/filter the data.

Upload screenshot: forward plotQualityProfile PDF

Choose File Screenshot …t 9.58.26 AM



Where does quality begin to noticeably decline in the forward reads?

220 bp

Chosen forward truncation length:

220 bp

1B) Reverse read quality profile

What is usually different about reverse-read quality compared with forward-read quality?

Reverse-read quality is typically lower and contains greater error rates because there is greater chemical decay and phasing.

Upload screenshot: reverse plotQualityProfile PDF

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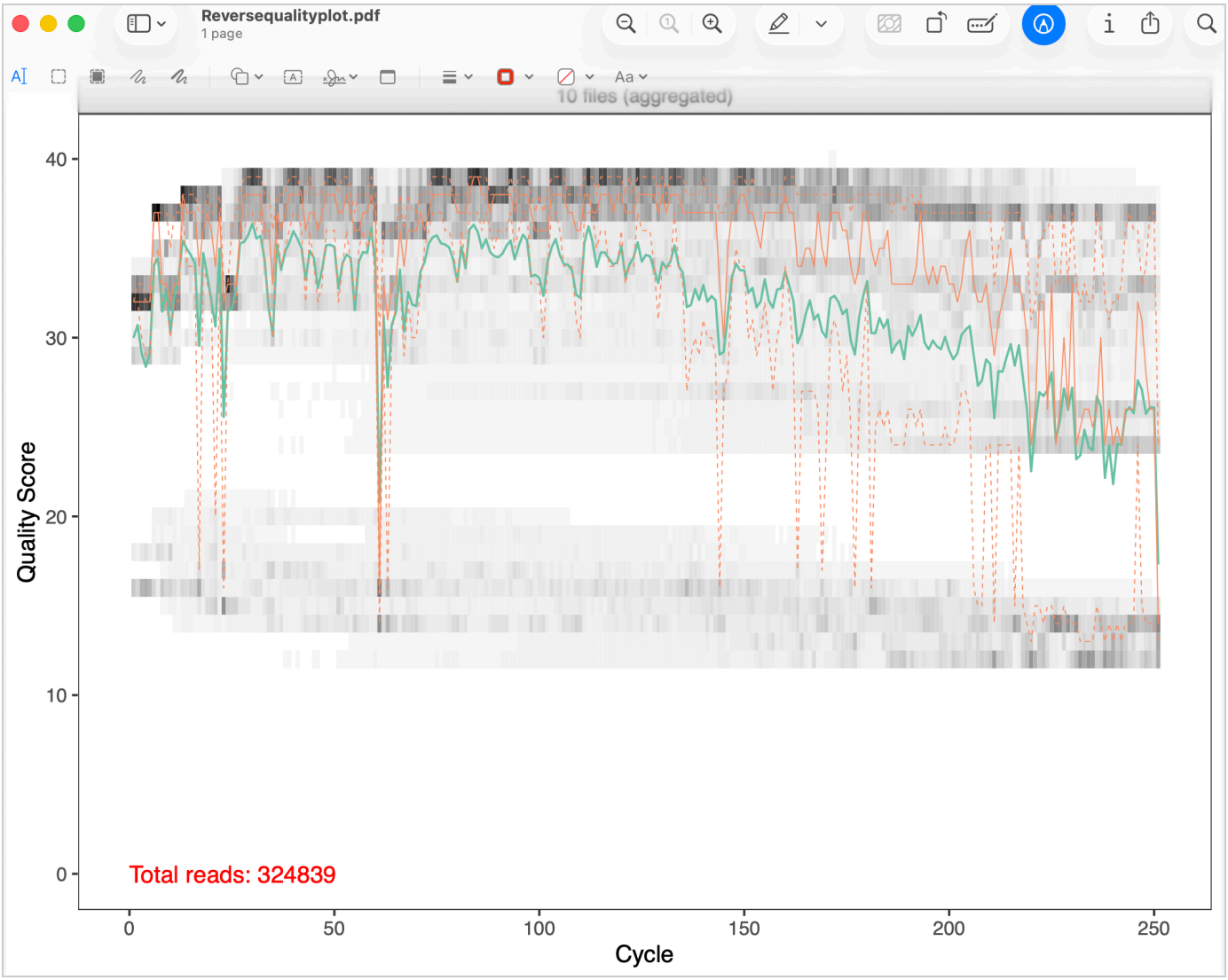

Where does quality begin to noticeably decline in the reverse reads?

205 bp

Chosen reverse truncation length:

205 bp

1C) Truncation decision

Explain why you chose these truncation values.

I chose the truncation value of 220 bp because that is where the first significant drop occurs in the forward quality plot. I chose the truncation value of 205 bp because this is where the first major drop occurs in the reverse quality plot. Even though the reverse quality plot is messy toward the beginning, the first big drop occurs

Rule of thumb: trim where quality begins to drop, but still keep enough overlap for forward and reverse reads to merge.

Part 2 — filterAndTrim

What filterAndTrim does:
This step removes low-quality reads and trims poor-quality bases from the ends of reads before denoising.

Why is this step important?

This step is important because it helps to prevent sequencing artifacts (low-quality bases and adapter sequences), which helps to further prevent alignment errors downstream. Therefore, it assists in improving the

Parameters used:

Under the separate filters and trimmers for reverse reads, I changed the setting from the default of no to yes.

What could happen if you trim too aggressively?

Trimming too aggressively can result in significant data loss and can decrease the read coverage, which can eventually impact genome assembly because high-quality data can be thrown out.

What could happen if you do not trim enough?

Not trimming the quality reads enough can lead to the sequence still containing contaminants, which can allow errors to remain in the dataset, which can significantly influence downstream analysis.

IMPORTANT — Split Paired Reads (Unzip Collection)

After `filterAndTrim`, your output is a **paired collection**.
Before continuing, you must split this into **forward and reverse collections**.

What to do in Galaxy

1. In the Tools panel, search: **unzip collection**
2. Input: your `filterAndTrim` paired output
3. Run

What you will get

- One collection = **forward reads (R1)**
- One collection = **reverse reads (R2)**

Why this matters

`learnErrors` and `dada` must be run **separately** on forward and reverse reads.
If you skip this, `mergePairs` will fail.

Part 3 — learnErrors

What learnErrors does:
This step models the sequencing error rates in your dataset so DADA2 can distinguish real sequence variants from sequencing mistakes.

Why is this step important?

This is an important step because we want to remove and correct reads that contain errors, helping to prevent a misidentification of our species.

Upload screenshot: learnErrors PDF

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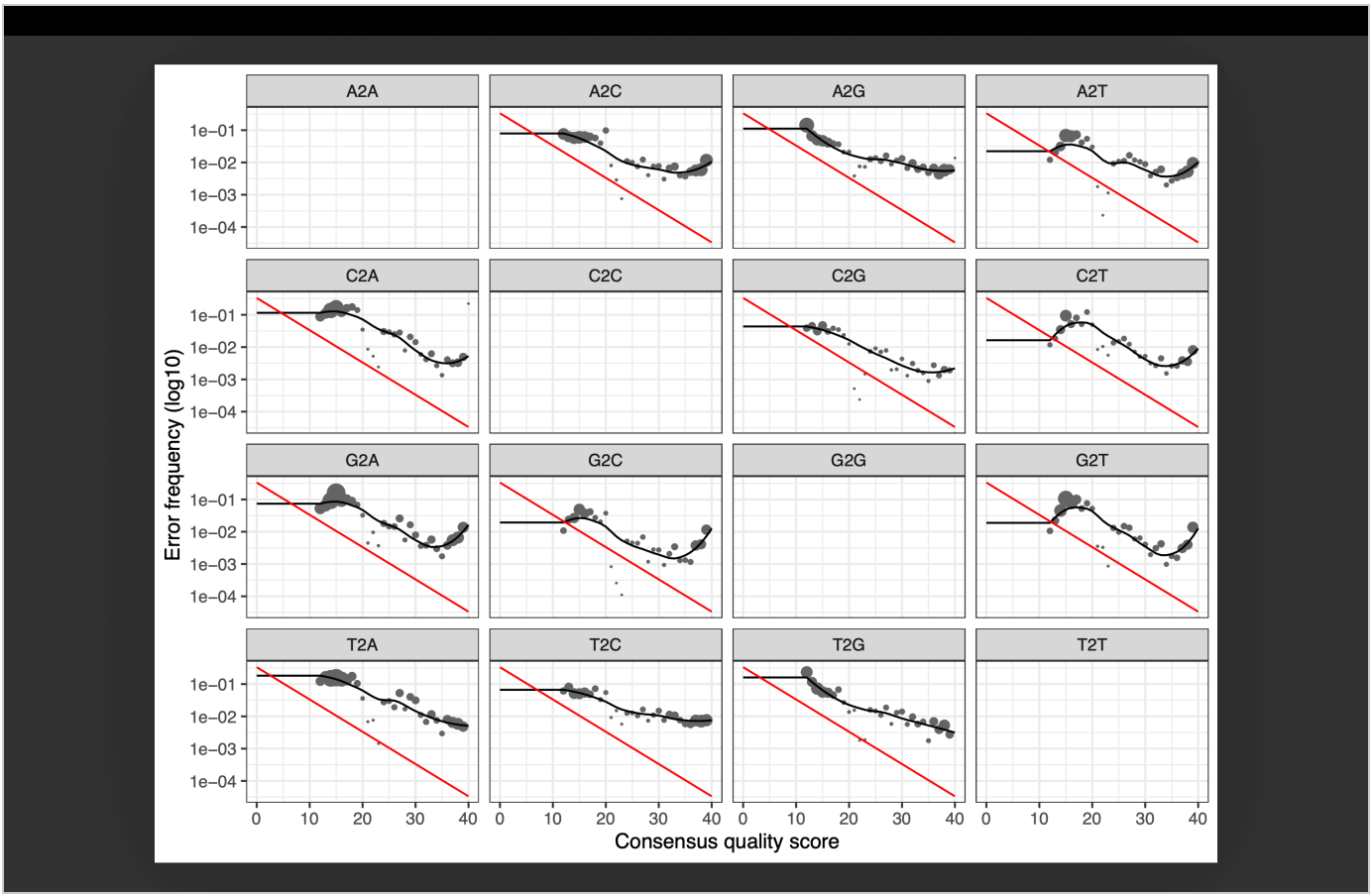

Does the fitted model generally follow the observed error pattern?

The fitted model generally follows the observed error pattern

Why is learning error rates important before denoising?

Learning error rates is important before denoising because it helps to prevent false positives, which would lead to an inaccurate assessment of the sequence. Instead of relying on default sequencing runs, it can more accurately infer sample composition (helps to remove errors from the sequencing).

Part 4 — dada (denoising)

What dada does:
This is the denoising step. DADA2 uses the learned error model to correct likely sequencing errors and infer exact amplicon sequence variants (ASVs).

What is denoising doing?

Denoising is the process of correcting errors in sequencing data to distinguish actual biological sequences from noise. It helps to detect errors in PCR and sequencing, hence reducing the number of errors and allowing for the

How is ASV inference different from OTU clustering?

clustering combines similar sequences into one unit, which could potentially combine multiple species. OTU clustering clusters sequences based on percentage similarity, while ASVs utilize statistical models to estimate the actual, error-free sequences. OTU clustering is more reliant on the dataset, while ASVs allow for more direct

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Part 5 — mergePairs

What mergePairs does:
This step merges the forward and reverse reads for each amplicon so you can reconstruct the full sequence from overlapping read pairs.

What is being merged, and why?

The forward and reverse ASVs are being merged together into a contig. The purpose of merging pairs is to reject pairs that do not overlap with one another or have mismatches. Thus, it helps to differentiate sequencing errors

Why do forward and reverse reads need to overlap?

Forward and reverse reads need to overlap because it confirms the accuracy of the sequence, reduces the number of errors, and makes a contig from a DNA fragment.

If reads fail to merge, what is one likely reason?

Reads can fail to merge because there is not good overlap between the paired-end reads, or because there are high rates of mismatches. Aggressive trimming and low quality raw data can lead to failed merged reads.

Part 6 — makeSequenceTable

What makeSequenceTable does:
This step creates the ASV table. Each row is a sample, each column is an ASV, and each cell contains the number of times that ASV was observed in that sample.

What does this table represent biologically?

Biologically, this table represents each ASV in the columns and each sample in the rows, which demonstrates the number of times each true biological sequence was observed. The exact variants are represented.

What are the rows, columns, and cell values in the sequence table?

The rows represent the different samples, the columns represent the ASVs, and the values represent the integer count of the sequence reads.

Part 7 — removeBimeraDenovo

What removeBimeraDenovo does:
This step removes chimeras, which are artificial sequences created during PCR when fragments from different templates are incorrectly joined together.

What is a chimera, and why should it be removed?

not completely amplified, which results in a left fragment from one parent, and a right fragment from the second parent. Therefore, it is a sequence that does not exist. Chimeras should be removed because they appear to be unique sequences, when actually they are falsely identified ASVs. Therefore, to improve the accuracy of the

Why are chimeras a problem for interpretation?

Chimeras are a problem for interpretation because they lead to false interpretations of diversity, and also lead to inaccurate abundances.

Part 8 — assignTaxonomy

What assignTaxonomy does:
This step compares your ASVs to a reference database and assigns taxonomy, such as phylum, class, order, family, genus, or sometimes species.

What does taxonomy assignment add to your ASV table?

Taxonomy assignment adds biological meaning to my ASV table because it maps the individual DNA sequences to previously known taxonomic ranks via reference databases. Therefore, it allows for the interpretation of the

Why might some ASVs not be assigned all the way to species?

Some ASVs might not be assigned all the way to species because they might be understudied and may not be in the reference databases. Also, if the sequencing reads are too short, then there might not be enough variable

Part 9 — Sequence counts summary

What sequence counts shows:
This output tracks how many reads remain after each major step in the pipeline so you can see where reads were retained or lost.

Why is it useful to track reads across steps?

Explain what this summary helps you evaluate.

Upload screenshot: sequence counts output

Choose File No file chosen

Approximately how many reads did you start with?

e.g., 45,000 per sample or similar

Approximately how many reads remained at the end?

e.g., 31,000 per sample or similar

At which step did you lose the most reads?

e.g., filtering, merging, chimera removal

What is one likely explanation for that loss?

Part 10 — Moving into phyloseq

After completing DADA2 in Galaxy, we can create a phyloseq object for analysis.

10A. Which files will you take from DADA2 from Galaxy to use in phyloseq?

I will be inputting the removebimeradenovo table, the assign taxonomy table, and the metadata sample table.

10B. Why must sample names in the metadata match sample names in the ASV table?

The sample names in the metadata must match the sample names in the ASV table to ensure that the software can link the data to the sample traits. This helps to connect the biological data contextually and proceed with

10C. In 2–3 sentences, explain how DADA2 transforms raw reads into data that can be analyzed ecologically in phyloseq.

DADA2 transforms raw reads into data that can be analyzed ecologically in phyloseq because it filters and trims the raw reads, differentiates biological variation from noise, then makes a table of ASVs. The paired-end reads are merged together, the chimeric sequence are removed, taxonomy is assigned to ASVs, and an accurate count is created. The ASV table is inputted into phyloseq for downstream analysis.

Export (PDF)

1. Click **Prepare submission preview** (generates a clean summary).
2. Click **Download as PDF (Print)** and save.

Submission preview

Dataset
Project/sample set: WCSP microbiome
Sequencing type: 16S V4 paired-end
Grouping variable: SampleSource

Quality profiles
Purpose: The purpose is to analyze the length of the quality of the sequence, specifically allowing us to see how where the quality of the sequence drops and visualize where to trim/filter the data.
Reverse vs forward quality: Reverse-read quality is typically lower and contains greater error rates because there is greater chemical decay and phasing.
Forward quality drop: 220 bp
Reverse quality drop: 205 bp
Forward truncation: 220 bp
Reverse truncation: 205 bp
Why these values: I chose the truncation value of 220 bp because that is where the first significant drop occurs in the forward quality plot. I chose the truncation value of 205 bp because this is where the first major drop occurs in the reverse quality plot. Even though the reverse quality plot is messy toward the beginning, the first big drop occurs at approximately 205 bp.

filterAndTrim
Purpose: This step is important because it helps to prevent sequencing artifacts (low-quality bases and adapter sequences), which helps to further prevent alignment errors downstream. Therefore, it assists in improving the accuracy, removing any contaminants, and increasing the alignment.
Parameters used:

```
Under the separate filters and trimmers for reverse reads, I changed the setting from
the default of no to yes.
```

If trimmed too much: Trimming too aggressively can result in significant data loss and can decrease the read coverage, which can eventually impact genome assembly because high-quality data can be thrown out.
If not trimmed enough: Not trimming the quality reads enough can lead to the sequence still containing contaminants, which can allow errors to remain in the dataset, which can significantly influence downstream analysis.

learnErrors
Purpose: This is an important step because we want to remove and correct reads that contain errors, helping to prevent a misidentification of our species.
Model fit: The fitted model generally follows the observed error pattern
Why important: Learning error rates is important before denoising because it helps to prevent false positives, which would lead to an inaccurate assessment of the sequence. Instead of relying on default sequencing runs, it can more accurately infer sample composition (helps to remove errors from the sequencing).

dada + mergePairs
Denoising purpose: Denoising is the process of correcting errors in sequencing data to distinguish actual biological sequences from noise. It helps to detect errors in PCR and sequencing, hence reducing the number

of errors and allowing for the true recognition of the sequence.
ASVs vs OTUs: ASV inference is different from OTU clustering because ASV produces strain-level resolution, while OTU clustering combines similar sequences into one unit, which could potentially combine multiple species. OTU clustering clusters sequences based on percentage similarity, while ASVs utilize statistical models to estimate the actual, error-free sequences. OTU clustering is more reliant on the dataset, while ASVs allow for more direct comparisons across species.
mergePairs purpose: The forward and reverse ASVs are being merged together into a contig. The purpose of merging pairs is to reject pairs that do not overlap with one another or have mismatches. Thus, it helps to differentiate sequencing errors from biological variations and improves the length of the sequence.
Need for overlap: Forward and reverse reads need to overlap because it confirms the accuracy of the sequence, reduces the number of errors, and makes a contig from a DNA fragment.
One reason reads fail to merge: Reads can fail to merge because there is not good overlap between the paired-end reads, or because there are high rates of mismatches. Aggressive trimming and low quality raw data can lead to failed merged reads.

Sequence table + chimera removal + taxonomy
Sequence table purpose: Biologically, this table represents each ASV in the columns and each sample in the rows, which demonstrates the number of times each true biological sequence was observed. The exact variants are represented.
Rows/columns/cells: The rows represent the different samples, the columns represent the ASVs, and the values represent the integer count of the sequence reads.
Chimera removal purpose: A chimera is a sequence of artificial DNA that is created during PCR when the parent sequence templates are not completely amplified, which results in a left fragment from one parent, and a right fragment from the second parent. Therefore, it is a sequence that does not exist. Chimeras should be removed because they appear to be unique sequences, when actually they are falsely identified ASVs. Therefore, to improve the accuracy of the downstream analysis and avoids false interpretations.
Why chimeras are a problem: Chimeras are a problem for interpretation because they lead to false interpretations of diversity, and also lead to inaccurate abundances.
Taxonomy purpose: Taxonomy assignment adds biological meaning to my ASV table because it maps the individual DNA sequences to previously known taxonomic ranks via reference databases. Therefore, it allows for the interpretation of the samples.
Taxonomy limits: Some ASVs might not be assigned all the way to species because they might be understudied and may not be in the reference databases. Also, if the sequencing reads are too short, then there might not be enough variable regions to differentiate between species that are closely related.

Sequence counts
Purpose: -
Starting reads: -
Ending reads: -
Largest loss step: -
Likely explanation: -

Move to phyloseq
Files to download:

```
I will be inputting the removebimeradenovo table, the assign taxonomy table, and the
metadata sample table.
```

Why names must match: The sample names in the metadata must match the sample names in the ASV table to ensure that the software can link the data to the sample traits. This helps to connect the biological data contextually and proceed with downstream analysis.
DADA2 to phyloseq transition: DADA2 transforms raw reads into data that can be analyzed ecologically in phyloseq because it filters and trims the raw reads, differentiates biological variation from noise, then makes a table of ASVs. The paired-end reads are merged together, the chimeric sequence are removed, taxonomy is assigned to ASVs, and an accurate count is created. The ASV table is inputted into phyloseq for downstream analysis.